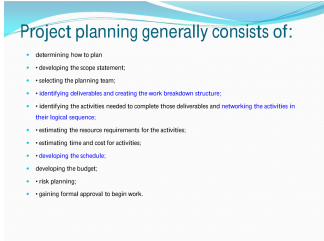
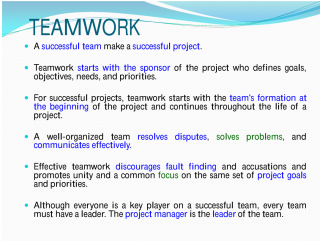
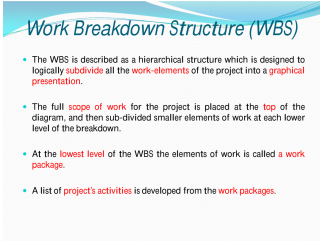
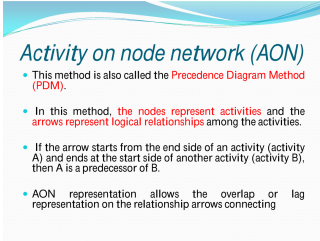


CONSTRUCTION AND PROJECT MANAGEMENT - I

Section B - Lecture 03

PROJECT PLANNING

Question-to-slide mapping for the 2016-2024 term-final question bank. Original Section A/B placement is preserved, but classification is based on the actual lecture content.

			
Planning process	Project teams	WBS	AON network

25 Mapped sub-questions Across nine examination years	315 Linked examination marks Includes partial-coverage questions	12 Exact matches 132 linked marks	2 Direct matches 27 linked marks	11 Partial matches 156 linked marks
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Core finding: The strongest recurring areas are WBS/OBS linkage, project-planning purpose and steps, project teams, and AOA/AON network construction. CPM/PERT calculations are only partially supported.

1. Coverage Summary and Repeated Patterns

Year	Linked marks	Relative coverage
2016	42	
2017	28	
2018	20	
2019	41	
2020	40	
2021	44	
2022	35	
2023	34	
2024	31	

Recurring theme	Main slides	Observed pattern
WBS, OBS and their linkage	L29-36	Directly tested in 2016, 2017, 2018, 2019, 2021 and 2023.
Planning meaning, purpose and procedure	L10-13, L25-28	Appears in 2016, 2019, 2020, 2021, 2022 and 2023.
Project teams and overlapping teamwork	L14-24	A major 15-mark question in 2022 and a useful support area for project-success questions.
AOA/AON network drawing	L47-64	Appears throughout the question bank, with a fully focused 20-mark AOA/AON question in 2024.
SMART objectives / constraints	L4, L9	SMART was asked in 2021 and 2024; constraint coverage is only partial.
Dummy activity	L52-53	Explicitly tested in 2019 and embedded in several AOA network problems.

Interpretation rule: "Exact" means the lecture substantially contains the requested answer. "Direct" means it teaches the same method with different data. "Partial" means only one component of a multi-part question is covered.

2.1 Detailed Question-to-Slide Mapping: 2016-2018

Year	Original Sec./Q	Marks	Condensed question	Lecture slide(s)	Match	Coverage assessment
2016	A 1(a)	10	Define construction management and discuss the characteristics of a project plan.	L10-13, L27-28	Partial	Planning purpose, contents and steps are covered; the construction-management definition belongs to another lecture.
2016	A 2(c)	25	Draw an AOA network, then calculate paths, critical path, schedules and float.	L47-53	Partial	The lecture supports network construction and dummy activities, but not CPM calculations, floats or schedule analysis.
2016	A 4(d)	7	Define WBS and OBS and show how they are linked with an example.	L29-36	Exact	Definitions, hierarchy, responsibilities and two linkage diagrams are provided.
2017	A 2(b)	20	Draw the project network and prepare ES, EF, LS, LF, float and critical-path analysis.	L47-53	Partial	Only the network-diagram portion appears in this lecture; time calculations are outside its scope.
2017	A 3(b)	8	Define WBS and OBS and demonstrate their linkage in a project.	L29-36	Exact	The WBS, OBS and responsibility-control-account link are directly illustrated.
2018	B 5(b)	8	Explain the time-cost-scope triple constraint and its effect on a project.	L4, L9	Partial	The planning diagram shows time and cost around performance goals, and SMART supports objectives, but the full constraint theory is not developed.
2018	B 7(b)	12	Define WBS and OBS and explain how a WBS is transformed into a network diagram.	L27-38, L47-60	Exact	The lecture moves from planning steps to WBS/OBS, activities, precedence and AOA/AON network construction.

QB = original question-bank PDF page. Slide numbers refer to the uploaded 64-page Project Planning PDF.

2.2 Detailed Question-to-Slide Mapping: 2019-2021

Year	Original Sec./Q	Marks	Condensed question	Lecture slide(s)	Match	Coverage assessment
2019	A 1(c)	8	Why do projects need planning, and what does planning involve for project success?	L10-13, L18, L25-28	Exact	Planning contents, purposes, consequences of weak planning and the planning workflow are explicit.
2019	A 3(a)	8	Transform a WBS into a network diagram and explain why WBS and OBS are linked.	L27-38, L47-60	Exact	The complete chain from WBS/OBS to activities, precedence and networks is shown.
2019	A 3(c)	15	Infer relationships among steelwork activities and draw a project network.	L37-42, L47-60	Direct	Activity classification, precedence questions and comparable worked examples support the task, although the exact steelwork data are not solved.
2019	A 4(a)	10	Define floats, dummy activity, optimistic time and pessimistic time.	L52-53	Partial	Dummy activity and its use are covered; floats and PERT time estimates are not.
2020	A 1(a)	8	Define a project and state the characteristics of a project and project plan.	L10-13	Partial	The project-plan portion is supported; the definition and general project characteristics belong to the introductory lecture.
2020	A 2(b)	20	Draw the network, identify critical path and prepare float calculations.	L38-60	Partial	The lecture supports precedence and network drawing but not CPM arithmetic.
2020	A 3(c)	12	Draw the network for a construction project containing real and dummy activities.	L47-53	Exact	AOA construction rules, dummy activities and corrected representations directly match the task.
2021	A 1(a)	12	Differentiate CM and PM, and state the SMART rule for project objectives.	L9	Partial	SMART is stated in full; the CM-PM comparison is not part of this lecture.
2021	A 3(a)	13	Define project planning and explain planning challenges using the Engineering V concept.	L11-13, L25-28	Exact	Planning meaning, purpose, the Engineering V problem-resolution diagram and planning steps are included.
2021	A 3(c)	12	Draw an AON network for a listed construction project.	L54-60, L63-64	Direct	AON rules, patterns, comparisons and worked exercises teach the method; the exact exam dataset is not reproduced.
2021	A 4(b)	7	Define WBS and OBS and show their linkage with an example.	L29-36	Exact	A direct repeat of the lecture definitions and linkage diagrams.

QB = original question-bank PDF page. Slide numbers refer to the uploaded 64-page Project Planning PDF.

2.3 Detailed Question-to-Slide Mapping: 2022-2024

Year	Original Sec./Q	Marks	Condensed question	Lecture slide(s)	Match	Coverage assessment
2022	A 3(a)	10	What is project planning, and what does project planning generally consist of?	L10-13, L25-28	Exact	The lecture gives both the general list and a numbered planning procedure.
2022	A 3(b)	15	Define project team and teamwork; explain working with multiple teams in an overlapping environment.	L14-24	Exact	Team definition, team building, teamwork, multiple teams, hierarchy and overlapping environment are fully developed.
2022	A 3(c)	10	Define WBS, motivation, eustress and distress.	L29-32	Partial	WBS is covered in detail; motivation is only briefly mentioned and stress concepts are in another lecture.
2023	A 2(b)	12	Write short notes on WBS, OBS, AOA networks and project planning.	L10-13, L29-36, L47-53	Exact	All four requested short-note topics are core parts of the lecture.
2023	A 3(b)	22	Draw an AON network and complete critical-path and schedule calculations.	L54-60	Partial	AON drawing is covered, but paths, ES/EF/LS/LF, float and duration analysis are not.
2024	A 1(a)	11	Define a project using the three constraints and discuss the SMART rule.	L4, L9	Partial	SMART is exact and the planning diagram references time and cost; the full three-constraint definition is not presented.
2024	A 3(b)	20	Draw both AOA and AON network diagrams from predecessor information.	L47-60	Exact	The lecture explicitly teaches both methods, their conventions, dummy rules, patterns and comparisons.

QB = original question-bank PDF page. Slide numbers refer to the uploaded 64-page Project Planning PDF.

3. Slide-to-Question Reverse Index

Slides	Core topic	Mapped questions	Priority
L4, L9	Planning constraints diagram and SMART objectives	2018 B5(b); 2021 A1(a); 2024 A1(a)	High
L10-13	Planning contents, meaning, purpose and consequences	2016 A1(a); 2019 A1(c); 2020 A1(a); 2021 A3(a); 2022 A3(a); 2023 A2(b)	Very high
L14-24	Project teams, team building, teamwork and multiple-team overlap	2022 A3(b)	Focused high-value
L25-28	Engineering V, planning inputs/outputs and numbered planning steps	2019 A1(c), A3(a); 2021 A3(a); 2022 A3(a)	Very high
L29-36	WBS, OBS, hierarchy, resources and linkage	2016 A4(d); 2017 A3(b); 2018 B7(b); 2019 A3(a); 2021 A4(b); 2022 A3(c); 2023 A2(b)	Highest
L37-46	Activities, precedence, bridge and WBS/OBS worked examples	2018 B7(b); 2019 A3(a), A3(c); 2020 A2(b)	High
L47-53	Network definition, AOA rules and dummy activities	2016 A2(c); 2017 A2(b); 2018 B7(b); 2019 A3(a), A4(a); 2020 A2(b), A3(c); 2023 A2(b); 2024 A3(b)	Highest
L54-60	AON rules, AOA-AON comparison and examples	2018 B7(b); 2019 A3(a), A3(c); 2021 A3(c); 2023 A3(b); 2024 A3(b)	Highest
L61-64	Additional relationship and network exercises	2021 A3(c); 2024 A3(b)	Practice

Fastest revision route: L29-36 -> L47-60 -> L10-13 -> L25-28 -> L14-24 -> L37-46 -> L61-64 -> L4 and L9.

4. Exam-Focused Answer Construction Guide

1. Project planning definition and purpose

Use L11 for the definition and three questions. Add L12 purposes, L10 general contents, and L13 consequences of inadequate planning.

2. Engineering V / planning challenges

Draw the Engineering V from L25. Explain movement from needs and functional requirements to solution design, construction, integration, testing and handover. Add inputs/outputs from L26.

3. WBS and OBS linkage

Define WBS from L29 and levels from L30-32. Define OBS from L33-34. Reproduce either linkage diagram from L35-36 and explain responsibility assignment.

4. Transforming WBS into a network

Follow L27-28: scope -> WBS -> OBS -> relationships -> estimates -> network. Then use L37-38 to identify activities and predecessors before drawing AOA/AON.

5. Project team and teamwork

Define project team from L14, team building and motivation from L15, teamwork from L18, then explain multiple-team overlap using L20-24 and the hierarchy figure.

6. AOA network

Use L47-51 for network rules. Explain arrow=activity and node=event. Add unique i-j numbering, no back arrows, unified start/end and dummy activity from L52-53.

7. AON network

Use L54-55: nodes represent activities and arrows represent relationships. Apply the AOA-AON comparisons in L56-58 and practise with L59-64.

8. SMART objective

Write Specific, Measurable, Acceptable, Realistic and Time-based exactly as L9. Add one short construction example for each term.

5. Coverage Limits and Final Study Priority

Question area	Lecture support	What is still needed
CPM calculations and floats	Network drawing, precedence and AOA/AON conventions	Critical-path arithmetic, forward/backward pass, ES/EF/LS/LF and float formulas require the CPM lecture.
PERT optimistic/pessimistic times	No substantive coverage	Use the PERT lecture for expected time, variance and probability calculations.
Triple constraint	Time and cost appear in the planning framework; SMART is complete	A formal explanation of scope-time-cost interaction is better taken from the introductory lecture.
Project definition and characteristics	Project-plan characteristics and planning steps only	Use the introductory lecture for the definition and general characteristics of a project.
Motivation and stress	Motivation is briefly mentioned in team building	Eustress, distress and project-manager psychology require the administration/psychology lecture.
Exact exam network datasets	Rules and comparable worked examples are provided	Several questions require constructing a new diagram from the supplied data; the lecture does not contain every final answer.

Study order	Slides	Focus
Priority 1	L29-36	WBS, OBS and linkage - the most repeatedly tested theory block
Priority 2	L47-60	AOA/AON networks, rules, dummy activities and comparisons
Priority 3	L10-13 and L25-28	Planning definition, contents, purpose, Engineering V and steps
Priority 4	L14-24	Project teams, teamwork and overlapping environment
Priority 5	L37-46 and L61-64	Activity relationships and worked network practice
Priority 6	L4 and L9	Planning constraints diagram and SMART objectives

Bottom line: This lecture is a primary source for planning, WBS/OBS, teamwork and network-drawing questions. It should be studied together with the CPM/PERT lecture whenever calculations are requested.